Stochastic Modeling Notes

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Chapter 1

Case Study: "Interbolsa"

1.1 Executive Summary

This case analyzes the collapse of InterBolsa S.A., the largest brokerage firm in Colombia in 2012, from the perspective of financial risk management and corporate governance. The study identifies how the combination of excessive leverage through repo operations, high concentration in an illiquid asset (Fabricato), structural dependence on short-term funding, and governance failures led to a systemic liquidity crisis.

The analysis demonstrates that internal risk models did generate meaningful early warning signals; however, those warnings were neither properly escalated nor incorporated into strategic decision-making. The progressive loss of confidence among banks, institutional investors, and market participants accelerated the deterioration until regulatory intervention became inevitable.

1.2 Business Context

During the first decade of the 21st century, Colombia experienced sustained growth in its capital markets, driven by economic expansion, increasing foreign investment, and institutional consolidation within the Colombian Stock Exchange (BVC).

Within this environment, InterBolsa became the leading brokerage firm in the country, expanding operations into Chile, Brazil, Panama, and the United States. Its business model evolved from traditional brokerage services toward sophisticated leveraged operations, repos, and financial structuring.

The firm operated under an organizational culture focused on aggressive growth and return maximization, where risk management gradually became subordinated to commercial objectives.

1.3 Problem Statement

The central problem of this case is to evaluate how a highly leveraged structure, financed through short-term repo operations and concentrated in a single illiquid collateral asset, generated a liquidity spiral and loss of confidence that ultimately resulted in InterBolsa's operational collapse.

This analysis seeks to answer the following questions:

- Which risks were ignored or underestimated?
- How did concentration in Fabricato amplify systemic vulnerability?
- What governance and risk communication failures existed?
- Why were stress-testing results not incorporated into decision-making?

1.4 Data Description

1.4.1 Sources of Information

- Academic case study on InterBolsa (2026)
- Illustrative financial exhibits
- Public information from the Colombian Financial Superintendence
- Reconstructed pedagogical datasets

1.4.2 Variables Analyzed

1.5 Methodology

1.5.1 Conceptual Framework

This analysis applies financial risk management principles related to:

- Liquidity risk
- Concentration risk
- Market risk
- Systemic risk
- Corporate governance
- Stress testing

1.5.2 Repo Financing Model

Repo operations can be represented as:

$$\text{Obtained Liquidity} = \text{Collateral Value} \times (1 - h) \tag{1.1}$$

where:

- h represents the haircut applied to collateral.

As Fabricato's price declined and haircuts increased, InterBolsa's refinancing capacity deteriorated rapidly.

1.5.3 Stress Testing Framework

The CRO developed stress scenarios based on:

- Declines in collateral prices
- Reduction in bank funding lines
- Increased margin calls
- Simultaneous liquidity and confidence deterioration

The severe stress scenario assumed:

$$\Delta P_{Fabricato} \leq -50\% \tag{1.2}$$

and bank funding availability reduced to:

$$L_{available} = 40\% \tag{1.3}$$

The model projected a critical liquidity gap:

$$\text{Liquidity Gap} = -130 \text{ billion COP} \tag{1.4}$$

1.6 Case Analysis

1.6.1 Concentration Risk

InterBolsa maintained approximately 56% consolidated exposure to Fabricato, significantly exceeding internal prudential limits.

This concentration generated several vulnerabilities:

- Excessive dependence on a single collateral asset
- Limited liquidation capacity in stressed markets
- High sensitivity to price fluctuations
- Reputational contagion risk

The progressive decline in Fabricato simultaneously deteriorated:

- Collateral value
- Creditor confidence
- Refinancing capacity

1.6.2 Liquidity Risk

The business model depended heavily on continuous refinancing through short-term repos.

Critical indicators included:

The loss of confidence triggered a classic liquidity crisis cascade:

1. Collateral price decline
2. Margin calls

3. Withdrawal of bank funding
4. Forced asset sales
5. Additional price declines
6. Systemic confidence crisis

1.6.3 Corporate Governance Failures

One of the most important findings of the case was the information asymmetry between the risk management area and the Board of Directors.

While executive dashboards reported:

- Diversified portfolio
- Satisfactory liquidity
- Controlled risk exposure

the technical risk department observed:

- Critical concentration levels
- LCR below minimum thresholds
- Excessive use of funding lines
- Severe stress-test deficits

This reflects a structural failure in corporate governance and organizational risk culture.

1.6.4 Failures in Risk Management

The main failures identified include:

- Underestimation of liquidity risk
- Ignoring stress-testing results
- Excessive dependence on short-term funding

- Incentives aligned with growth rather than risk control
- Ineffective escalation mechanisms for the CRO
- Overconfidence in the investment thesis

1.7 Results and Findings

The case demonstrates that InterBolsa did not collapse solely because of financial losses, but due to a simultaneous crisis involving:

- Liquidity deterioration
- Loss of market confidence
- Weak corporate governance
- Excessive concentration risk

Internal models identified the collapse scenario months in advance; however, the organization lacked both the governance structure and cultural willingness to act upon those warning signals.

1.8 Risk Interpretation

From an integrated risk management perspective, the InterBolsa case highlights:

- The dangers of procyclical leverage
- The fragility of short-term funding models
- The critical importance of liquidity management
- The relevance of realistic stress testing
- The central role of confidence in financial markets

The case also illustrates that:

Liquidity disappears faster than solvency.

Financial crises typically develop gradually over time and then materialize abruptly.

1.9 Recommendations

- Implement strict concentration limits
- Strengthen CRO independence
- Establish binding stress-testing frameworks
- Improve transparency toward the Board of Directors
- Diversify funding sources
- Develop formal liquidity contingency plans
- Incorporate forward-looking supervisory metrics

1.10 Limitations

- The exhibits contain illustrative data
- Complete audited financial statements were unavailable
- Part of the analysis is reconstructive
- The case emphasizes pedagogical dimensions

1.11 Conclusion

The InterBolsa case represents one of the most important examples of systemic liquidity collapse in Latin American financial markets.

The firm did not fail exclusively because of losses associated with Fabricato, but because of the interaction between concentration risk, leverage, collateral deterioration, and the rapid disappearance of market confidence.

The principal lesson from this case is that quantitative models alone are insufficient if organizations lack a strong risk culture, effective governance mechanisms, and the willingness to act decisively on early warning signals.

1.12 References

Variable	**Description**
Total Repos	Financial leverage level
Fabricato Concentration	Exposure to a single issuer
LCR Proxy	Liquidity coverage ratio
Haircuts	Market confidence deterioration
Bank Credit Lines	Dependence on external funding
Fabricato Price	Collateral deterioration

Table 1.1: Key variables analyzed in the case

Indicator	**October 2012**
LCR Proxy	0.61x
Bank line utilization	97%
Average repo maturity	18 days
Average haircut	31%

Table 1.2: Critical liquidity indicators

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